

WebBartosz Portfolio — Architecture Analysis

Project: WebBartosz Portfolio
Technology: Vite, vanilla JavaScript, CSS3, HTML5
Location: /home/dominic/Documents/WebBartosz
Report #: 2/20

Architecture Type

Static single-page application

Layers

Static HTML/CSS/JS → Vite Build → Deployed to static host

Design Patterns

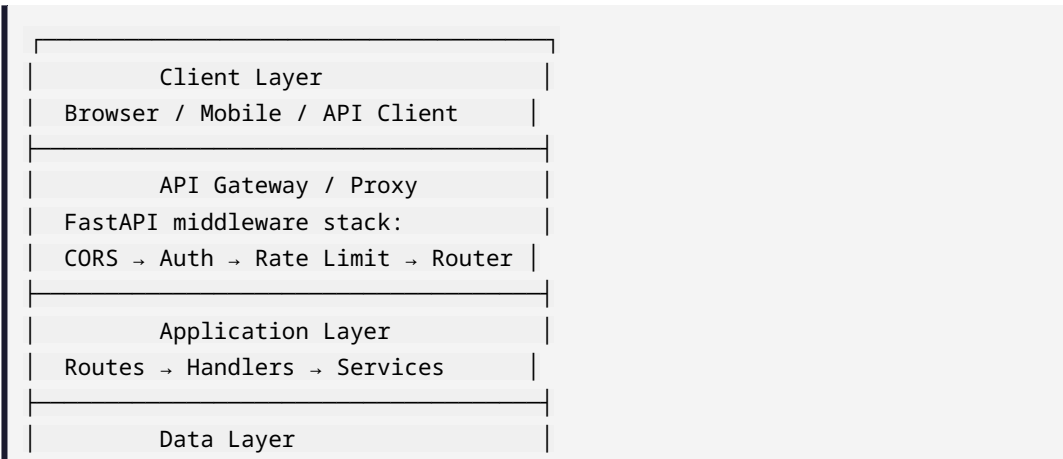
SPA with hash-based routing

Scalability

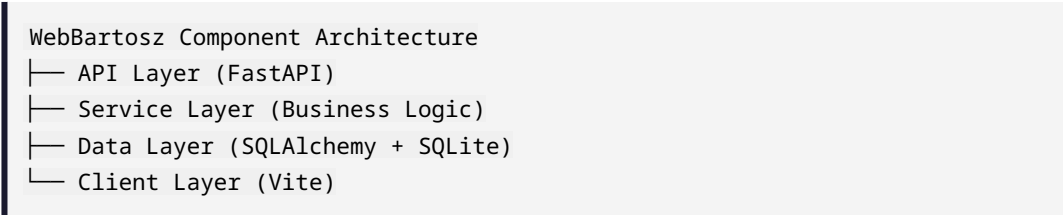
Static site, infinitely scalable via CDN

System Architecture

Layer Architecture



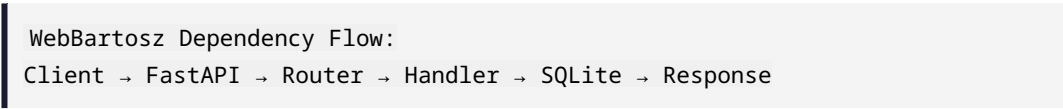
Component Interaction



Request Lifecycle

1. **HTTP Request** arrives at FastAPI
2. **CORS middleware** checks origin headers
3. **Auth dependency** extracts and validates JWT
4. **Rate limiter** checks request count per IP
5. **Router** matches path to handler
6. **Handler** executes business logic
7. **Database session** commits or rolls back
8. **Response** returned to client

Dependency Graph



Design Patterns Used

Pattern	Location	Purpose
Dependency Injection	FastAPI Depends()	DB sessions, auth
Repository Pattern	SQLAlchemy models	Data abstraction
Middleware Chain	FastAPI middleware	Cross-cutting concerns
Router Module	APIRouter include	Separation of concerns